

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GPH 414 — Past Questions & Materials
by Renance

Course: GPH 414
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 400

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Paper 1 of 4 — GPH 414 Geoph. Well-log & Form Evaluat

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

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DEPARTMENT OF EARTH APPLIED GEOPHYSICS

FIRST SEMESTER: B.Sc. (GEOPHYSICS) DEGREE EXAMINATION 2024/2025 SESSION

COURSE: GPH 414 (2 Units) GEOPH. WELL-LOG & FORM EVALUAT. TIME ALLOWED: 2" tirs.

INSTRUCTION: ANSWER THREE (3) QUESTIONS IN ALL, QUESTION ONE IS COMPULSORY THEN ANSWER

ONE QUESTION EACH FROM SECTION B ANDC

We

SECTION A

1. Describe a typical laboratory Practical session for drilling mud analysis. List the equipment used, the

key properties measured, and explained the importance of each test in drilling operations. (10 marks)

(b) What are the main purposes of casing in a wellbore? Explain the different types of casing strings used in drilling and the criteria for selecting casing sizes and grades. (7 marks)

(c) Outline the functions of drilling mud during well operations. Discuss how properties such as mud weight, viscosity, and filtration are controlled and why they are controlled and why they are important.

(6 marks)

) Differentiate water based mud and oil based mud. (2 marks)

7 Explain Logging while drilling (LWD) and measurement while drilling (MWD) and list types of LWD logs. (5 marks)

SECTION B |

2. (a) State one advantage and one limitation of the Hingle crossplot method. (4 marks)

(b) The data in the table below were extracted from logs of a sandstone reservoir of the Agbada Formation of the tertiary Niger Delta basin. Using the Hingle grid with $a = 1.0$, $n =$
 $A_t = 185 \mu s/t$,

(i) graph the data using the Hingle crossplot method and determine;

2.0 and $m = 2.0$;

(10 marks)

(ii) the interval transit time in the formation matrix (A_{tma}) at Ro-line infinite resistivity intercept. (2 marks)

(iii) the formation water resistivity (R_w).

(5 marks)

(iv) the water saturation (S_w) and the Hydrocarbon saturation (S_o) at a point with resistivity value of

26.00Om. (6 marks)

(v) the porosity of the point in (iv) above. (3 marks)

Zone Number Depth (ft) Atiog (Lis/ft) R, (Qm)

1 9,310 63.0 26.0

2 9,322 52.0 18.0

a 3 9,326 53.5 17.0

58.0 9.0

: 5396 oes 32.0

5 : 4.8

6 9,347 61.5

; 9.353 61.5 4.7

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5 9,367 52.5 22.0

10 9,373 53.5 11.0

11 9,376 57.5 5.1

12 9,383 62.0 3.3

13 9,387 62.0 3.3

14 9,398 58.0 6.4

15 9,400 55.0 5.1

NOTE: Graph sheets will be provided in the examination Hall.

3. List the methods of computing water saturation (S_w) known to you and discuss any two of the methods in details. (11 marks)

) Dual Induction Log and Focused Log data recorded through a shaly sand reservoir from the pay interval of the reservoir are given as; $R_{ip} = 70 \text{ Ohm-m}$, $R_{im} = 105 \text{ Ohm-m}$, and $R_e = 320 \text{ Ohm-m}$.

i) Using the Chart (Figure 1), determine d_w , R/R_{ILD} , R_w , R_{yo}/R ; and R_x . (6 marks)

ii) If $R_{ip} = 1.2 \text{ Ohm-m}$ at 70° F (21° C) and $R_y = 0.033 \text{ Ohm-m}$ at formation temperature 180° F (82° C)

using ratio method equation, determine S_y , S_w , and the hydrocarbon saturation S_{he} . (11 marks)

ip Indicate using results from (ii) above, if the hydrocarbon were moved during invasion. (2 marks)

SECTION C

4. (a) The Figure-2 below shows an SP and resistivity logs acquired over a formation in the Niger Delta.

What can you conclude from the fact that the resistivity logs overlay at some interval sand separate at

other intervals, also comment on why the shallow reading tool resistivity values is higher than those of

the mecium and the deep reading tools (i.e $R_{yo} > R_i > R_w$)? (8 marks)

(b) How does the presence of gas affect the following?

(i) Density log response

(ii) Neutron log response i

(iii) | Sonic log response? (6 marks)

(c) What effect is shale on the neutron log response? (3 marks)

(d) List three (3) uses of the Resistivity log. (Ginatk)

(a) (i) Delineate the top and bottom depths of the reservoirs A, B, and C in Figure-3 below. (3 marks)

(ii) What is the approximate thickness of the gas-bearing zone of reservoir A? (1 mark)

2

J

(iii) What are the approximate thicknesses of the oil-bearing zone of reservoirs A and B? (2 marks)

(iv) What would reservoir C be likely saturated with? State the reason for your assertion? (3 marks)

(v) At what depth is the GOC in reservoir A? (1 mark)

(b) Calculate the density porosity (O_p) at a depth of 1051ft of the gas bearing portion of reservoir A,

and if the true resistivity is 150Qm, compute the formation water saturation (S_w) in this interval, given

that the formation water resistivity (R_w) is 0.5Qm at formation temperature. What then is the gas saturation (S_g) in this interval? (Hint: $m = 2.15$, $n = 2$ and $a = 0.61$ in the Archie's equation and

$P_{ma} = 2.65 \text{ g/cm}^3$, $\rho_{py} = 1.0 \text{ g/cm}^3$). (10 marks)

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Figure-2: SP and resistivity logs acquired over a formation in the Niger Delta.

Figure 3: Log set taken in a vertical exploration well in Niger Delta.

Paper 2 of 4 — GPH 414 GEOPH WELL LOG FORMATION EVALUATION

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

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DEPARTMENT OF EARTH APPLIED GEOPHYSICS

B.Sc. (GEOPHYSICS) DEGREE EXAMINATION 2024/2025 SESSION

INSTRUCTION: ANSWER THREE (3

(3) QUESTIONS IN ALL, QUESTION ONE IS COMPULSORY THEN ANSWER ONE QUESTION EACH FROM SECTION B AND C

hy

SECTION A

1. (a) Describe a typical laboratory practical session for drilling mud analysis. List the equipment used, the

key properties measured, and explained the importance of each test in drilling operations. (10 marks)

(b) What are the main Purposes of casing in a wellbore? Explain the different types of Casing strings used in drilling and the criteria for selecting casing sizes and grades. (7 marks)

uch as mud

(c) Outline the functions of drilling mud during well operations. Discuss how properties such as weight, viscosity, and filtration are controlled and why they are controlled and why they are important.

(6 marks)

2. (a) Differentiate water based mud and oil based mud. (2 marks)

(b) Explain Logging while drilling (LWD) and measurement while drilling (MWD) and list types of LWD

logs. (5 marks)

: SECTION B

2. (a) State one advantage and one limitation of the Hingle crossplot method. (4 marks)

(b) The data in the table below were extracted from logs of a sandstone reservoir of the Agbada Formation of the tertiary Niger Delta basin. Using the Hingle grid with $a = 1.0$, $n = 2.0$ and $m = 2.0$; $A_t = 185 \mu\text{s/ft}$; |

(i) graph the data using the Hingle crossplot method and determine;

(ii) the interval transit time in the formation matrix (A_t

(iii) the formation water resistivity (R_w).

(10 marks)

(c) (i) at R_o -line infinite resistivity intercept. (2 marks)

(5 marks)

(iv) the water saturation (S_w) and the Hydrocarbon saturation (S_i) at a point with resistivity value of 26.00m. (6 marks)

(v) the porosity of the point in (iv) above. (3 marks)

Zone Number Depth (ft) Atiog (us/ft) R; (Qm)

1 9,310 63.0 26.0

| 2 9,322 52.0 18.0

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12 9,383 62.0 3.3

13 9,387 62.0 3.3

14 9,398 58.0 6.4]

15 9,400 55.0 5.1

NOTE: Graph sheets will be provided in the examination Hall.

w) Known to you and discuss any two of the

3. List the methods of computing water saturation (S

(11 marks)

methods in details.

Ab) Dual Induction Log and Focused Log data recorded through a shaly sand reservoir from the pay = 320 Ohm-m.

(6 marks)

interval of the reservoir are given as; $R_{ito} = 70$ Ohm-m, $R_{itm} = 105$ Ohm-m, and R_{et}

(if Using the Chart (Figure 1), determine d_j , $R/RILD$, R_i , $R_{.}/R$; and $R_{.}$

“(ii) If $R_{ing} = 1.2$ Ohm-m at $70^\circ F$ ($21^\circ C$) and $R_y = 0.033$ Ohm-m at formation temperature $180^\circ F$ ($82^\circ C$)

using ratio method equation, determine S_y , S_w and the hydrocarbon saturation S_{hc} . (11 marks)

fii) Indicate using results from (ii) above, if the hydrocarbon were moved during invasion. (2 marks)

SECTION C

4. (a) The Figure-2 below shows an SP and resistivity logs acquired over a formation in the Niger Delta.

What can you conclude from the fact that the resistivity logs overlay at some interval and separate at

other intervals, also comment on why the shallow reading tool resistivity values are higher than those of

the medium and the deep reading tools (i.e. $R_{sh} > R_{msl} > R_{deep}$)? (8 marks)

(b) How does the presence of gas affect the following?

(i) Density log response

(ii) Neutron log response

(iii) Sonic log response? (6 marks)

(c) What effect is shale on the neutron log response? (3 marks)

(d) List three (3) uses of the Resistivity log.

(3 marks)

5. (a) (i) Delineate the top and bottom depths of the reservoirs A, B, and C in Figure-

Figure 3 below. (3 marks)

(ii) What is the approximate thickness of the gas-bearing zone of reservoir A? (1 mark)

(1 mark)

2

(iii) What are the approximate thicknesses of the oil-bearing zone of reservoirs A and B? (2 marks)

(iv) What would reservoir C be likely saturated with? State the reason for your assertion? (3 marks)

(v) At what depth is the GOC in reservoir A? (1 mark)

(b) Calculate the density porosity (ϕ_p) at a depth of 1051 ft of the gas-bearing portion of reservoir A,

and if the true resistivity is 150 Ωm , compute the formation water saturation (S_w) in this interval, given

that the formation water resistivity (R_w) is 0.5 Ωm at formation temperature. What then is the gas saturation (S_g) in this interval? (Hint: $m = 2.15$, $n = 2$ and $a = 0.61$ in the Archie's equation and

$\rho_{ma} = 2.68 g/cm^3$, $\rho_{fl} = 1.0 g/cm^3$). (10 marks)

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Figure b: Dualtiduction StL Tormado chart (Schlumberger, 1998, Chart Rint-2c¢)

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Figure 3: Log set taken in a vertical exploration well in Niger Delta.

Paper 3 of 4 — GPH 414 Geophysical Well Logging and Form Evaluation.

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

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DEPARTMENT OF EARTH APPLIED GEOPHYSICS

FIRST SEMESTER:

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B.Sc. (GEOPHYSICS) DEGREE EXAMINATION 2024/2025 SESSION

COURSE: GPH 414 (2 Units) GEOPH, WELL-LOG & FORM EVALUATION. Time ALLOWED: 2hrs?

INSTRUCTION: ANSWER THREE (3) QUESTIONS IN ALL, QUESTION ONE IS COMPULSORY THEN ANSWER

ONE QUESTION EACH FROM SECTION B AND C

Hrs.

SECTION A

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(c) Outline the functions of drilling mud during well operations. Discuss how Properties such as weight, viscosity, and filtration are controlled and why they are controlled and why they are important.

1. equipment used, the

operations. (10 marks)

types of casing strings

(7 marks)

such as mud

important.-

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(d) Differentiate water based mud and oil based mud. (2 marks)

(e) Explain Logging while drilling (LWD) and measurement while drilling (MWD) and list types of LWD

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SECTION B

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(b) The data in the table below were extracted from logs of a sandstone reservoir of the Agbada Formation of the tertiary Niger Delta basin. Using the Hingle grid with $a = 1.0$, $n = 2.0$ and $m = 2.0$; $At = 185 \text{ ys/t}$:

(i) graph the data using the Hingle crossplot method and determine: (10 marks)

(ii) the interval transit time in the formation matrix (Δt_m) at R_{∞} -line infinite resistivity intercept. (2 marks)

(iii) the formation water resistivity (R_{w0}).

(5 marks)

(iv) the water saturation (S_{w0}) and the Hydrocarbon saturation (S_{h0}) at a point with resistivity value of 26.00m. (6 marks)

(v) the porosity of the point in (iv) above. (3 marks)

Zone Number Depth (ft) Δt_{log} (Ls/ft) R_t (Q_m)

1 9,310 63.0 26.0

2 9,322 52.0 18.0

3 9,326 63.5 17.0

4 9,332 | 58.0 9.0

5 9,335. | 52.5 32.0

6 9,347 61.5 4.8

7 9,353 61.5 4.7

8 9,362 53.0 40.0

9 9,367 52.5 22.0

10 9,373 53.5 11.0

11 9,376 | 57.5 5

12 9,383 62.0 3.3

13 9,387 62.0 3.3

14 9,398 58.0 6.4

15 9,400 55.0 5

NOTE: Graph sheets will be provided in the examination Hall,

- (a) List the methods of computing water saturation (S_{w0}) known to you and discuss any two of the methods in details.

(11 marks)

from the pay

70 Ohm-m, $R_{itp} = 105$ Ohm-m, and $R_e = 320$ Ohm-m.

(i) Using the Chart (Figure 1), determine d_i , R/R_{ILD} , R_y , R_{xofRt} and R_{∞} .

(b) Dual Induction Log and Focused Log data recorded through a shaly sand reservoir for interval of the reservoir are given as; $R_{itp} =$

(6 marks)

(it) If $R_{ag} = 1.2$ Ohm-m at 70° F (21°C) and $R_y = 0.6\%$ Ohm-m at formation temperature 180°F (82°C) -

using ratio method equation, determine S_w , S_{∞} and the hydrocarbon saturation S_{he} . (11 marks)

(ili) Indicate using results from (ii) above, if the hydrocarbon were moved during invasion. (2 marks)

SECTION C

resistivity logs acquired over a formation in the Niger Delta.

What can you conclude from the fact that the resistivity logs overlay at some interval and separate at

(a) The Figure-2 below shows an SP and

other intervals, also comment on why the shallow reading tool resistivity values are higher than those of

the medium and the deep reading tools (i.e. $R_{sh} > R_i > R_d$)? (8 marks)

(b) How does the presence of gas affect the following?

(i) Density log response

(ii) Neutron log response

(iii) Sonic log response? (5 marks)

(c) What effect is shale on the neutron log response? (3 marks)

(d) List three (3) uses of the Resistivity log. (3 marks)

(A) (i) Delineate the top and bottom depths of the reservoirs A, B, and C in Figure-3 below. (3 marks)

(ii) What is the approximate thickness of the gas-bearing zone of reservoir A? (1 mark)

2

(iii) What are the approximate thicknesses of the oil-bearing zone of reservoirs A and B? (2 marks)

(iv) What would reservoir C be likely saturated with? State the reason for your assertion? (3 marks)

(v) At what depth is the GOC in reservoir A? (1 mark) -

(b) Calculate the density porosity (G_p) at a depth of 1051ft of the gas bearing portion of reservoir A,

and if the true resistivity is 150Ωm, compute the formation water saturation (S_w) in this interval, given

that the formation water resistivity (R_w) is 0.50Ωm at formation temperature. What then is the gas saturation (S_g) in this interval? (Hint: $m = 2.15$, $n = 2$ and $a =$

0.61 in the Archie's equation and

$P_{ma} = 2.65 \text{ g/cm}^3$, $p_f = 1.0 \text{ g/cm}^3$).

(10 marks)

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Figure 1: Dual Induction-SFL Tornado chart (Schlumberger, 1998, Chart Rint-2c).

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Figure-2: SP and resistivity logs acquired over a formation in the Niger Delta.

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Paper 4 of 4 — Well logging GPH 414

Level 400 • First Semester • Handout

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3. Subsurface Facies Analysis

Douglas J. Cant, Geological Survey of Canada, 3303 - 33rd St. N.W.
Calgary, Alberta T2L 2A7

INTRODUCTION DIFFERENCES FROM data from geophysical logs, 3 to

This chapter will attempt to bridge the SURFACE WORK 20 cm-diameter cores, and cross section methodological and scale gap between Subsurface data provide a differentially tions constructed from them, cannot

sedimentology based largely on out- biased sample of a rock unit than match the level of local detail available

crops and modern environments, and outcrop data. Drill holes and cores are from an outcrop. Seismic data give a

techniques and results of large-scale, concentrated in localities and zones of view of a basin on a very much larger

subsurface investigations. It is an intro- economic interest whereas outcrops scale than outcrop studies. The differ-

duction to 1) geophysical logs and cor- preferentially expose rocks which are ences in the appropriate scales of

relation, 2) subsurface facies analysis, harder and more resistant to weath- investigation for outcrop compared

3) seismic methods, and 4) larger- ering. Because drill holes and geo- with subsurface studies are extremely

scale analysis of sedimentary facies physical logs normally sample a con- significant, both scientifically and

successions and allostratigraphy. tinuous, uninterrupted section (where- economically.

Subsurface work lends itself to the as outcrops rarely do), subsurface cor-

study of facies relationships on a scale relation is based on more complete Subsurface methods larger than can be accomplished on data. Seismic-reflection data may The following sections will briefly

most outcrops. Consequently, many provide a coherent three-dimensional review the principles behind sub-

relatively new ideas concerning allo- picture of a basin, along with informa- surface methods of investigation, in-

stratigraphy, sequence stratigraphy, tion on the relationships between the cluding both geological methods (well

base-level changes, and global strati- sediments and structural features of a logs and cores), and geophysical

graphic correlations have emerged basin. methods (seismic-reflection data).

!from subsurface geology, and are Subsurface data cannot provide as Other publications discuss these

based on both geological and geo- much local facies information. No mat- techniques in more detail, notably

physical data. ter how closely spaced wells may be, Krumbein and Sloss (1963), Allen

Table 1 Log types, properties measured, and geological uses.

Log Property Measured Units Geological Uses

Spontaneous Natural electric potential (compared to Millivolts Lithology (in some cases), correlation,

potential drilling mud) curve shape analysis, identification of porous zones.

Resistivity Resistance to electric current flow Ohm-metres Identification of coals, bentonites, fluid evaluation.

Gamma-ray Natural radioactivity — related to API units Lithology (shaliness), correlation, curve K, Th, U shape analysis.

Sonic Velocity of compressional sound wave Microseconds/metre Identification of porous zones, coal,

tightly cemented zones.

Caliper Size of hole Centimetres Evaluate hole conditions and reliability of other logs.

Neutron Concentrations of hydrogen (water and Per cent porosity Identification of porous zones, cross

hydrocarbons) in pores plots with sonic, density logs for empirical separation of lithologies.

Density Bulk density Kilograms per cubic Identification of some lithologies

(electron density) includes pore fluid metre (gm/cm³) such as anhydrite, halite, nonporous in measurement carbonates.

Dipmeter Orientation of dipping surfaces by Degrees Structural analysis, stratigraphic resistivity changes (and direction) analysis

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(1975), Payton (1977), Selley (1978), can be used to define a "shale line" saturated However, where low-perme-

Anstey (1982), and Miall (1984). Some (Fig. 1). Deflections of the log from the ability rocks occur, such as the tightly

newly developed methods in subsur- shale line indicate zones of permeable cemented sandstones of the western

face sedimentology, such as the use rock containing interstitial fluid with Alberta Basin or the bitumen-saturated

of resistivity microscanners to detect salinities different from the drilling Athabasca Oil Sands, the SP log cannot -

sedimentary structures, and the anal- mud. reliably distinguish lithologies If subsur-

ysis of high-resolution dipmeter data, Experience in many basins has face formations contain fresh rather than

are as yet not widely applied, but are shown that the SP log may be a good indicator of saline water (such as some Upper

discussed by Hurst et al. (1990). The indicator of lithology in areas where sand- Cretaceous rocks of Alberta), SP deflection

two major types of data, subsurface stones are permeable and water saturation is suppressed or even reversed

geological and geophysical, will be treated separately, but where available, both can be integrated.

GEOLOGICAL USES OF WELL LOGS

Well logs are made by pulling an instrumented tool up the hole, and recording the data as a function of depth. They are used extensively in the petroleum industry for evaluation of the fluids in rocks, but this aspect will not be covered here. The interested reader is referred to the various logging company manuals, or to Asquith (1982). Geophysical logs are the fundamental source of data in many subsurface studies because virtually every oil and gas well is logged from near the top to the bottom. Almost all well logging is done by pulling the measurement tool up the hole on the end of a wire. Different types of logs and the properties they measure are shown in Table 1 and are discussed briefly below. On any well log, the absolute elevation of any bed or bed contact is obtained by subtracting its depth in the well from the surveyed elevation of the Kelly Bushing (KB) on the drilling platform; this elevation is given on the top (header) of the well log.

Spontaneous potential (SP) log

This log records the electric potential between an electrode pulled up the hole and a reference electrode at the

surface. This potential exists because of electrochemical differences between the waters within the formation and the drilling mud, and because of ionic selection effects in shales (the surfaces of clay minerals selectively allow passage of cations compared to anions). The potential is measured in millivolts on a relative scale only. A shale line is shown in Figure 1 Example of SP and resistivity logs from the Alberta Basin. Two resistivity logs are shown — one of medium depth (resistivity) and another (deep induction) which reads farther into the rock beyond the influence of the drilling mud. The deep induction log shows lower resistivity (Fig. 1) because the absolute value in porous zones, probably indicating salt-water saturation. The carbonates are Devonian depends on the properties of the Winterburn dolomites overlain by Cretaceous Mannville Group sandstones and shales. The drilling mud. In shaly sections, the curved arrows indicate individual facies successions; these can be seen as progressive maximum SP response to the right upward deflections in both the SP and resistivity logs.

3. SUBSURFACE FACIES ANALYSIS 29

from normal, depending on the salinity horizontal distance from the well bore. rated porous rocks have high resistivities, so the log can be used in these cases to separate shales from porous lithology is to calibrate the log against length of the tool and focusing the sandstones. Several resistivity and Gamma-ray log surface during normal drilling by the on the same track (Fig. 1). This log (Fig. 2) measures the natural gamma-ray emission of circulating mud) and hence gain experience in a particular area. Resistivity logs are used for evaluation of fluids within formations. They can also be used for identification of elements related to their content of radioisotopes of potassium, uranium and thorium. These elements (particulate matter) are common in clay.

current, either transmitted directly to in TaBte"-T. In older wells where few minerals and some evaporites. In ter-

the rock through an electrode, or mag- types of logs were run, the resistivity rigenous clastic successions the" log re-

netically induced deeper into the for- log may be useful for picking tops and flects the "cleanness" (lack of clays) or

mation from the hole (induction logs; bottoms of formalions, and tor^coTre^ "phgjiness" fhigh radioactivities on the

Fig. 1). The term "deep" here refers to lating between wells. Freshwater-satu- AP|_scale, Fig. 2) of the rock, averaged

over an interval of about 2 m. Because

11-3-70-11W6 of this property, gamma-ray log pat-

terns mimic v e r t i r a l sandsCQOlgpJ nr

carbonate-content trends of fades suc-

cessions. It must be emr 1 '

the gamma-ray reading is •'.•<•

tion of grain size or carbo; <-: -

only of the proportion o\ faiMOcViviC

ements, which may be { - e ; f e j IrA^C.

proportion of shale. For example, ciay-

free sandstones or conglomerates with

"any mix of sand and pebble clast sizes

generally give similar responses, and

lime mudstone gives the same re-

sponse as grainstone. Distinguishing

between clean (clay-free) lithologies

such as sandstones, conglomerates,

dolomites, and limestones is best done

by calibration of one or more lqqsJo__

jc^s^icujtings,

Once the main lithologies are known,

Figure 3 Relationship between gamma-ray deflection and proportion of shale. A

Figure 2 Gamma-ray, caliper, and sonic logs from the Alberta Basin. Because of space cutoff half-way between maximum and

limitations, coaly shales (higher gamma-ray readings) are labelled coals. The lower section minimum values corresponds to about 28

consists of the Ireton and Leduc Formations and the upper section shows regressive per cent shale, and is commonly used as a

shoreline successions of the Upper Mannville Spirit River Formation. criterion for lithologic mapping.

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the gamma-ray log can be calibrated to problems with the gamma-ray log, 1) interconnected porespace]

lithology by establishing minimum and the log response may be affected by type of pores. The log is

maximum readings corresponding to diagenetic, radioactive clays in pores, useful for delineating beds of low-velocity

pure carbonate or sandstone versus 2) shales rich in illite (high K) are micaceous material such as coal (Fig. 2) or

pure shale end members. The concentration of radioactive materials in montmorillonite poorly cemented sandstones, as well

as the concentration of radioactive elements in shale and clonites or chlorite, and 3) as high-velocity materials such as

increases with compaction so the sandstones (high-velocity) are, tightly cemented sandstones and car-

bonate shale lines should be readily identified. If a more radioactive than those lacking fission products or igneous basement. Sonic

thick section is being studied. The tool feldspar. Calibration of the log against logs are also important in understanding

response is nonlinear (Fig. 3), so a core or cuttings may be necessary to log and calibrating seismic lines, as

cut-off halfway between the shale line distinguish lithologies in some cases. explained below.

and sandstone (or other clean lith-

ology) line has a value of roughly Sonic log Porosity logs

30 per cent shale for purposes such as This log (Fig. 2) measures the velocity Density and neutron logs can be derived -

as lithologic mapping. of sound waves in rock. This velocity played as estimates of porosities. The There are three main interpretation depends on 1) lithology, 2) amount of density tool emits gamma radiation

which is scattered back to the detector

in amounts proportional to the electron

density of the rock. The electron density,

in most cases, is related to the density of

the solid material, and the amount and

density of pore fluids. Density porosity is

calculated by assuming a density of the

solid material (2650 kg/m³ for sandstone

and shale, 2710 kg/m³ for limestone) and

fluid (1146 kg/m³ for salt water).

The neutron log measures the hydrogen

concentration (in water or

petroleum) in the rock. The tool emits

neutrons of a known energy level, then

measures the energy of neutrons re-

flected from the rock. Because energy is transferred most easily to particles of similar mass to the hydrogen concentration can be estimated. Neutron porosities are calculated by assuming that oil or water fills the pore spaces. Gas, or water bound into clay, mineral, or anomalous low values.

Caliper log

This log records the diameter of the hole (Fig. 2), and gives an indication of its condition and hence the reliability of other logs. A very large hole indicates that dissolution, caving or falling in of the rock wall has occurred, which can lead to errors in log responses. This log is particularly useful in mixed evaporite successions where dissolution has preferentially leached out soluble evaporites. A hole smaller than the drill bit size may be present because the fluid fraction of the drilling mud invades permeable zones, leaving the solid fraction (mud or filter

Figure 4 A gamma-ray cross section in the Upper Mannville Group of Alberta illustrating cake) on the inside of the hole. In one

correlation by pattern matching. The correlations have been made using the following criteria, 1) facies successions do not show abrupt lateral changes in character, 2) facies successions do not show abrupt changes in thickness, 3) facies successions do not show Alberta, very permeable, matrix-free

seaward (right to left) coarsening, 4) correlated surfaces slope seaward (to the left); conglomerate can be recognized on the

Coals (blank) were lost. 0" sonar logs are not spaced proportionally to Mc caliper log where the note size is similar to distances between wells. than the bit size.

3. SUBSURFACE FACIES ANALYSIS 31

Dipmeter log techniques; Griffiths and Bakke, 1990) cause they are es-

This log is made by a resistivity tool may be the primary tools in the future. pes. In the Alberta with three or four electrodes mounted. At present, most geologists match log Basin sh Scale Horizonji is a

on separate arms with a common patterns by eye (or by tracing and shale rich in organic debris. It can be

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3. SUBSURFACE FACIES ANALYSIS 31

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centre point. The orientation of the tool overlaying logs), allowing for variations identified over most of the basin by its

is also continuously recorded. Where in lithologies, thicknesses, and com- characteristic high-gamma ray, slightly

a dipping bed is encountered, the completeness of section. Three major cor- high-resistivity, and high-porosity

3) for the lithologic change takes relation methods will be discussed, 1) values. It is used in many studies

place at different elevations on each marker beds, 2) pattern matching, and the units above and below it (particu-

a) The direction and magnitude of 3) slice techniques. Early the Viking Formation), as a horizontal datum for cross sections. The

Marker beds at the top of the Middle Mannville Bluesky

The dip meter

measures structural The log response ("kick") of a distinct Sandstone is also a prominent marker

but can also detect various types of bed or series of beds can be used in the Mannville Group. The Bluesky is

of sedimentary dips such as a common as a marker (e.g., Fig. 19 in Chapter a shoreline deposit abruptly overlain

draped over, a steeply sloping 12) even if the lithology or origin of the by marine shales; this contact is a

or even some cross-stratified bed is not known. Distinctive, laterally bounding discontinuity used to define

lithology. In many cases it is difficult to extensive groups of beds commonly allostratigraphic units. determine the nature of a dipping result from transgressions or regressions. In other situations, markers are not

surface unless a core has been cut. sions or erosional episodes which related to allostratigraphic units. For

distribute proximal sediment far across example, volcanogenic bentonites are

CORRELATION OF LOGS the basin. Markers that can be easily recognized on logs (Table 1), Correct correlation of stratigraphic mapped regionally may therefore be providing reliable markers as well as

units is absolutely necessary to make related to, or include, important al- time lines. reliable cross sections and maps, and allostratigraphic surfaces. For example,

to conduct regional facies analysis. condensed sections (possibly ex- Pattern matching

Complex numerical procedures for pressing into a single log This technique involves recognition

matching and correlation of logs (such Chapter 1) are perhaps the most and matching of distinctive log patterns

as a method adapted from gene-typing extensive marker beds, and are ex- of any origin. The correlated patterns

may represent vertical facies successions (Fig. 4), superimposed on

Successions, or unconformity-bounded units (Fig. 19 in Chapter 12). The surfaces of the units chosen may be transgressive and separate individual facies successions (Fig. 4). Alternatively, surfaces of maximum transgression separate individual units (to be discussed in more detail below). The bounding surfaces of the chosen units may also be unconformities, such as Cardium Formation surface E5 (Fig. 19 in Chapter 12).

By matching patterns, correlations are made on the basis of log shapes over intervals of metres or tens of metres, rather than on individual peaks, troughs, or markers within the succession. Pattern matching may allow correlation even with lateral variations in lithologies, facies, and thicknesses of units have occurred (Fig. 4). In difficult cases, correlation is facilitated by tracing one log and overlaying it on an adjacent log. The logs can be moved up and down until the best overall fit is

Figure 5 Gamma-ray cross section correlated by pattern matching which shows facies obtained. Constantly changing posi-

successions sloping seaward and downlapping against a lower surface. Correct pattern matches of fit may indicate lateral facies

matching as shown in Figure 5. D indicates downlap PL thickness changes, and may indicate

and T indicates toplap. The logs are not spaced proportionally to distances between wells. Note synsedimentary tectonism. ...

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Pattern matching is extremely useful the other methods discussed above. from a large number of wells. Correlation

because it can be used to correlate The thicknesses of slices should be justified by data bases and lateral

facies successions or allostratigraphic chosen to minimize complications. For survey systems has progressed to the

units as defined from cores or outcrop. It example, if sandstones in JNs\ area point where automatic map production therefore facilitates investigation of re- average 30 m in thickness, the choice of of some attributes is dorae routinely gional facies relationships. Examples of slices less than 30 m may not yield inter- and quickly. this are shown in Figure 15 of Chapter pretable results. In some cases, trial and Subsurface geological maps are 12, and Figure 16 of Chapter 9. Detailed error is necessary to find the optimum either compilations of data, or i<nterp_re^___ pattern matching of individual facies solution. tive summaries, DatiTcompTTation maps successions may delineate surfaces of Slice techniques are most commonly of many different quantities have been toplap, downlap or onlap (Fig. 5), and agplied jn nonmarine deposits. Here, made, but for stratigraphic and sedi- can therefore be used to define large- other techniques do not work well mentological purposes, there are three scale or composite allostratigraphic because of the lack of continuous beds main types, 1) structure maps that. units. Pattern matching has worked ex- and absence of laterally extensive facies show the elevation of a surface (the tremej^well irTmanv studies oTshaitow- successions. By noting the stratigraphic example in Fig. 20 of Chapter 12 is manne_sediments in the Cretaceous position and thickness of each lithologic similar to a structure map), 2) isopach Western Interior Seaway of North unit with respect to a marker (commonly maps that show the thickness oTVunTT America, largely because of the lateral the top of the unit), the data is in a form (Fig. 21 in Chapter 9), and 3) litholog- uniformity of these rocks (at least com- of maximum utility when computerized. ical maps that show the composition of . paTelTRolrier facies). Thicknesses of slices can be changed a unit in one of several way_sJFig, 23 in Certain assumptions about the styles easily until patterns emerge. Note that CTiapter 9). Examples include maps of of lateral variation of facies must be slice techniques may produce correla- the thickness of carbonate rocks as a made in pattern matching in order to tions which cut across depositional units percentage of the total thickness, maps decide between alternative possibilities. or unconformities if they are applied to of net sandstone thickness, or maps of For example, the cross sections of units with sloping depositional surfaces. the ratio of clastic to carbonate thick- Figures 4 and 5 show that in these Flach (1984) has used this technique to nesses. Full descriptions of the different shoreline-to-shallow-marine elastics, the map sandstone-filled channels in the map types are given by Krumbein and

correlated units satisfy a number of con- nonmarine to estuarine McMurray Sloss (1963) and Miall (1984). Inter-
 ditions: 1) individual facies successions Formation of Alberta. He was able to take comparative summary maps and block diagrams do not show abrupt changes in thicknesses of such aspects as facies distributions, log patterns, or inferred grain reservoirs at different stratigraphic levels even though precise correlation could be achieved.
 size patterns; 2) individual successions meant supply directions are also not be achieved.
 do not show seaward increases in the commonly made (Figs. 18, 21, 22 in thicknesses of sandstones; 3) successions slope gradually seaward (unless SUBSURFACE MAPS Note that all maps depend absolutely on synsedimentary or postsedimentary tectonic Mapping in the subsurface differs little on correct correlation of units. If correlationism has occurred); and 4) abrupt from surface work except for the huge lateral changes in facies successions volume of data which can be collected will be worthless. imply that an unconformity exists.

Readers may be able to make different correlations through the same wells, depending in part on the initial assumptions made.

Figure 6. Slicing technique.

As a method of last resort, when no other method yields results, an interval can be subdivided by arbitrarily slicing it either into units of constant thickness, or into units with thicknesses proportional to the entire interval (Fig. 6). Slicing an interval does not give true correlations; it is only a way of splitting a section which cannot be subdivided any other way.

The implicit assumption is made that time lines through the interval are essentially horizontal. Where this assumption is invalid, slice techniques may yield

Figure 6. Slicing technique. (a) gamma-ray log from the Upper results which are grossly in error. It is a Cretaceous Horseshoe Canyon Formation, Alberta. The section between the E

means of last resort, but is necessary in Formation and the basal marker was subdivided into six equal slices, chosen to include but

some situations which do not yield to not split the major channel sandstones. Simplified from Nurkowski and Rahmani (1984).

3. SUBSURFACE FACIES ANALYSIS 33

SUBSURFACE FACIES ANALYSIS Log curve shapes ment. It follows that interpretation

Without cores, the definition and inter- The shapes of well-log curves have based on log curve shape alone is ex-

pretation of facies in the subsurface long been interpreted in terms of de- tremely imprecise. In those specific

are generalized and imprecise. Making positional facies because of their studies where log patterns have been

lithological logs of cores is similar to resemblance to grain size succes- calibrated to well-understood deposi-

the measurement of outcrop sections, sions (e.g., Selley, 1978). Where SP- tional facies successions in cores and/

the most obvious differences being Resistivity or Gamma-Sonic pairs of or outcrops, the log-pattern method

the scale of features which can logs are used, the patterns are mir- can be applied successfully to the in-

be observed in a limited-diameter rored, resulting in (among others) Jjelk. terpretation of correlative uncored

(commonly 9 cm) core, and the lack of s j i a p e d a n d funnel-shaped profiles facies successions.

oriented sedimentary structures for pa- (notelheHfunnel in the - lower TialTof The scale of facies successions is

leocurrent analysis, iahates^ and mud- well 7-10-62-7W5 in Figure 15 of also a very important criterion con-

stones.however, are commonly more Chapter 12). Much published work straining the interpretation of curve

easily studied in cores than in out- uses a simplistic "pigeon-hole" ap- shapes. For example, funnel-shaped

crops (the trace fauna is particularly proach to interpretation. An example is patterns (Fig. 7) may range from a few

well displayed in cores, Chapter 4). the classification of bell-shaped metres to several hundred metres

Cores should always be examined gamma-ray profiles as fining-upward. in thickness. These are appropriate

with the logs present to check for com- meandering-stream facies succes- scales (respectively) for a crevasse.

pleteness of recovery, core-log corre- sions. Problems with this approach will (\$Pla\$ building into an TnterdisTnbutaiy;

lation, and log responses. Well-log be discussed below. _B|^j[Fig. 5 in Chapter 9), andjLEiQ-

cross sections (Fig. 15 in Chapter 12; The most typical vertical patterns grading deltaic succession (Figs. 7, 16

Fig. 19 in Chapter 9) and appropriate seen on gamma-ray, SP and resistivity In Chapte7~9~V kinds of maps can extend the interpre- logs are shown in Figure 7. It is em- Difficulties in interpretation of log tations made from cores, and set them phasized in this diagram that no patterns may result from deviations of into a larger-scale stratigraphic and pattern is unique to, or diagnostic of, individual facies successions from the paleogeographic context. any particular depositional environ- general model in Figure 7, possibly in Figure 7 The most common idealized log curve shapes which may be interpreted by correlation with many different core examples (see other chapters). Where resistivity or sonic logs are shown opposite gamma-ray logs, the patterns are mirrored, resulting in the bell and funnel labels. The limitations of this approach to facies analysis, in the absence of other data, is emphasized in the text, and by the multiple possible environmental interpretations indicated below each curve. Log curve shapes, in the absence of other data, are not diagnostic of particular environments.

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some cases because of base-level Condensed sections a result of cut-off of clastic supply. In changes. Other difficulties result from These stratigraphic intervals represent carbonate successions, the sediment- the amalgamation of units. For ex- periods of vrrv low sedimentation producing environments are drowned. ample, the standard clastic prograding rates in marine depositional environ- Condensed sections may show up lith- shoreline succession (left side of ments as a result of major transgres- ologically in carbonate (and some Fig. 14 in Chapter 12) generates sions (Fig. 27 in Chapter 13). They clastic) successions as hardgroundt. a continuously upward-broadening commonly characterize maximum with early diagenetic carbonate ana funnel-shaped gamma-ray pattern flooding surfaces, and can also be phosphatic cementation (see Chapter (Fig. 7). Sharp-based sandy shoreline present at marine flooding surfaces 4). In other clastic successions, they facies successions (right half of Fig. 14 (Chapter 1). Condensed sections in occur as intervals of oolite and lime in Chapter 12, also Fig. 15 in Chapter 12) clastic rocks are commonly formed as mud sedimentation. In basinal shales may result from small drops in relative sea level followed by progradation (forced regression, Chapter 12), or erosional transgression followed by shallow water regressive deposition

(Chapter 12). In both of these cases, the typical funnel-shaped log pattern is replaced by a cylindrical or blocky profile (Fig. 15 in Chapter 12), probably not correctly interpretable from log pattern alone.

Amalgamation of facies successions can also alter the standard log profiles of Figure 7. Superimposed channel deposits of meandering rivers can generate a stacked sandstone body characterized by a cylindrical log profile. Similarly, a transgressive sandstone capped by a regressive shoreline sandstone may be characterized by a cylindrical profile and conceal a transgressive unconformity or ravinement (Chapter 12). In the Middle Mannville Bluesky Formation, shoreline sandstones are in some places superimposed directly on nonmarine sandstones in the overall transgressive succession. These form sharp-based, cylindrical log patterns. These examples illustrate that simplistic labelling of log curve shapes in terms of depositional environments, without core or outcrop information, should be avoided.

RECOGNITION OF BOUNDING

DISCONTINUITIES

Allostratigraphic units are defined by their bounding discontinuities (Chapter 1). Because subsurface data can be used to document lateral relationships on a large scale, recognition of these discontinuities (maximum flooding surfaces, surfaces of marine transgression and regressive surfaces of erosion) has become extremely important in subsurface correlation by pattern matching. The correlation lines slope seaward (to the left), outlining cli-

face investigations. Delineation of these noform surfaces which tangentially downlap onto transgressive shallow-marine shales. The

discontinuities allows stratigraphic sub- section is hunn r.n mo mn nf tho Middie Maor.ville Elussky Formption (Datum). V>?

division of rocks into large-scale genetic maximum Hooding surface is the top of the transgressive ^haie (about 2a m above DHI.

packages, with possible implications for The sandstones (shaded) were deposited on the original clinoform slope. The logs are not

large-scale facies relationships. spaced proportionally to distances between wells.

3. SUBSURFACE FACIES ANALYSIS 35

¥ V

V the condensed horizons may be organic-rich radioactive shales, or pelagic deposits such as chinks or Seismic Stratigraphy section below), well-log cross sections (Fig. 8), and a few large outcrops. Clinoforms are developed in response to transgressive deepening followed by the re-establishment of laterally prograding sedimentary layers. In Figure 8, regressive shelf-to-shoreline clastic facies successions slope seaward creating a marine shales and limestones deposited in lagoons and estuaries as a result of transgression.

Unconformities

Surfaces of erosion or bypass are generally identified where underlying markers or facies successions are truncated (Fig. 19 in Chapter 12), or

W chalky shales. Because of these litho- clinoform, and terminate against, overlying ones show onlapping rela-

W logic contrasts with adjacent rocks, as or downlap onto the transgressive tionships. Well-log cross sections of

well as their wide lateral extent in shales and sandstones below. The marine rocks commonly allow defini-

many basins, condensed sections may condensed section (and in this case, tion of very low-angle, regional uncon-

be recognized on logs as marker beds the maximum flooding surface) lies di- formities which may be undetectable

with characteristically different well-log rectly below the surface of downlap. In on the scale of most outcrops. The

some nonmarine units, surfaces similar to condensed sections may be identified. A good example is the Fish Scale Horizon, which is commonly used as a horizontal datum for regional cross sections in the Cretaceous of the Alberta Basin. of extensive marker beds or easily correlated facies successions. As shown in Figure 5, detailed correlations of shoreline facies successions may be recognized on seismic lines (see show surfaces of onlap and downlap, hence defining minor bounding discontinuities.

In other examples, unconformities may be inferred in shoreline and nonmarine sections where locally distinctive stratigraphic units or successions recognizable locally are cut out and replaced by anomalous units. An example is shown in the cross section (Fig. 9) from the Lower Cretaceous Mannville group of eastern Alberta. Here, the gamma-ray logs in wells 11-30-55-14W4 and 6-32-55-13W4 can be correlated in detail by matching coarsening-upward and fining-upward patterns in these shoreline and nonmarine sediments (interpretations from numerous cores). Sonic and resistivity logs (not shown) were also used to make the best possible pattern matches. The thick sandstones just below the datum in wells 7-33, 7-36 and 6-31 are anomalous deposits

which locally replace the interbedded sandstones and shales. Numerous cores and some outcrops show that these thick sandstones are fluvial to estuarine in origin. Mapping has shown that they are linear bodies extending for tens of kilometres along the basin, but generally less than 5 to 8 km wide.

Figure 9 Gamma-ray cross section from the Upper Mannville of eastern Alberta. The Reasonable correlations can be made

regionally correlative log patterns in these fluvio-deltaic deposits (shown at each end) are between the sediments on either side

replaced locally by the valley-fill sandstone. Mapping has shown these anomalous gamma- of the channel (Fig. 9, wells 11-30 and

ray log patterns occur in a linear trend. Detailed pattern matching across the sandstones 6-32), probably indicating that layers

implies later incision of valleys into the top of the unit, rather than a contemporaneous were once continuous, but have subse-channel.

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Figure 10 The seismic image of an unconformity (or any other surface) depends on the acoustic impedance across it. The upper diagram shows rock units with their sonic velocities (km/sec). The lower diagram shows reflections, with amplitudes indicated by the thicknesses. The polarity of the reflection (i.e., whether a peak or a trough is generated first) depends on the impedance change, whether positive or negative. Where the polarity is reversed, the unconformity is indicated by a wavy line. The shaded peak of the reflection actually occurs one-half wavelength below the unconformity here.

6 - 33 - 14 - 13W4

Figure 11 Well 6-33-14-13W4, in southern Alberta, showing (right) a gamma ray log, (top left) a display of sonic log data from the well, seismic data from shot points 194 and 196, with the synthetic seismogram between them, and (bottom

left) a seismic line through the well. The tops of the Mannville Group (MNVL), Middle Mannville Ostracod Zone (OCDZ), and Mississippian (MSSP) are marked on each. The shot points (SP) are indicated above the seismic line. The peak (to the right) of each reflection is infilled. Note the lack of vertical resolution of the seismic data compared to the well log. However, the seismic line shows the context of the lower Mannville sandstone in a valley cut on the sub-Cretaceous unconformity. Note the truncations of reflections at the sub-Cretaceous unconformity. Modified from a well-integrated subsurface geology and seismic study by Hopkins et al. (1987).

SUBSURFACE FACIES ANALYSIS 37

requently been incised by valleys. A integration of facies analysis performed on the magnetic record entirely re-

series of correlative incision events at on cores with allostratigraphic analysis reflected energy. By moving the point of

the same stratigraphic level marks an essentially done on well-log cross section - origin of the sound waves (the shot

unconformity which is undetectable tions. By itself, core logging could not point) and the detectors, a continuous

where the channels are absent. These have shown the relationships between section of any desired length may be

anomalous sandstone units were the estuarine deposits and the older obtained. Seismic sections are nor-

therefore deposited within valleys cut deltaic sediments. Equally, allostratigraphically shot in straight lines.

as part of an unconformity that developed graphically analysis of cross sections A seismic reflection is generated

opened on fluvial and thin deltaic sediments - would not have yielded much information - where a descending sound wave en-

quences. The most generally accepted notion about the nature of the sediments counters an interface which reflects

interpretation of the series of events in and their relationships to sea level. part of the energy back to the detectors

this part of the Mannville Group (Fig. 9) tions. The reflectivity of a surface has

is: 1) deposition of the interbedded SEISMIC REFLECTIONS been found to depend on the contrast

fluvial and deltaic sands and shales, 2) Seismic surveys are conducted to in- in the density and sound transmission

drop of relative sea level, allowing the vestigate subsurface geology by velocity of the materials above and

incision of 30 m-deep fluvial valleys sending compressional sound waves below it. The product of these quanti-

through the coastal plain, 3) rise of into the earth and detecting the re- ties is the acoustic impedance of the

relative sea level (transgression) and flected or refracted energy returning medium, and the strength or amplitude

salt water invasion of the v a l l e y s , to the surface. Shallow (up to sev- of a reflection from a surface depends

forming linear estuaries, and 4) com- eral hundreds of metres penetration) on the acoustic impedance contrast or

plete filling of the valley by transgres- single-channel systems (one receiver) ratio, across it. Reflections are gener-

sive sediments grading from fluvial to or medium depth (up to 10 km pene- ated from stratigraphic surfaces such

estuarine deposits. tration) multi-channel systems (many as bed contacts or unconformities only

This interpretation illustrates the in- r e c e i v e r s , with d i g i t a l d a t a a d d e d where a contrast in acoustic i m p e d -

Figure 12 Seismic section from the Venture area, Scotian Shelf of eastern Canada. The horizontal scale is distance, but the vertical scale is

two-way travel time (up to 4 seconds), and is nonlinear with respect to depth in metres (the Venture B-13 well is 5219 m deep). The age data

and formation picks are from the two wells shown, tied in by synthetic seismograms. In this area, growth faults affect the Upper Jurassic to

Lower Cretaceous section, but younger rocks are almost undisturbed. Curved arrows indicate units of upward-increasing reflection frequen-

cies, amplitudes, and continuities. ESB indicates an erosional sequence boundary.

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ance occurs across them. For exam- mic section is not a geological cross commc -n at the tops of many

ple, an unconformity is not cL section. Individual sedimentary units sections ,ure 11, the 200 m-thick

imaged if sandstones of simila are not imaged, regardless . sir Mannvii • : oup is represented by locity and density are superimp lithology, if there is no impedan con- about truee peaks and troughs.

above and below it. However, latt trast at their surfaces. Also, ! er- Therefore the stratigraphic thickness

where the sandstone lies be; tical dimension of a seismic se^, n is represented by the average reflection

shale with lower velocity, a reflects , not depth, but the two-way travel time (peak to peak) is about 67 m. It is

generated (Fig. 10). (down and back) of the sonic wave therefore emphasized that one reflection does not represent a single bed,

A seismic section comprises a series of shot points displayed horizontally increase with greater depths, the but a stratigraphic interval several tens

metres in thickness. zontally, and a series of reflections display is nonlinear with respect to to hundreds of metres in

thickness. shown at different vertical distances. depth. Depth conversion can be done In many cases, depths are obtained

In Figure 11 (lower left), the shot roughly by using interval velocities, by using synthetic seismograms. Syn-

points are 25 m apart, and each shot calculated during data processing, and thetics are created by assuming or

point has a seismic trace displayed

beneath it. By convention, the right

half of each trace is filled in (black-

ened), for illustrative purposes. Where

the trace deflects markedly left or

right, a reflection from an interface is

indicated. Reflections are character-

ized by their amplitude and frequency,

and by the degree of continuity of

peaks and troughs from shot point to

shot point. A seismic section appears

very much like a geological cross sec-

tion (Fig. 12), with a series of surfaces

Figure 13 Stratigraphic patterns commonly seen on seismic lines, at many differer;

of contrasting acoustic impedance dis-

scales. When combined with biostratigraphic dating, these patterns are the basis for int-

played horizontally, appearing like

pretation of the stratigraphic history of a basin. Similar patterns but generally of a smai<-

sedimentary bedding. However, a seis- scale, may be identified on correctly correlated well-log cross sections.

1 - Alluvial conglomerate and sandstones.lacustrine shales 4 - Slope shales and interbedded sandstones

2 - Non-marine clastic and coal 5 - Deep marine clastic fans

2b - Deltaic facies 6 - Lagoonal-Shelf carbonates;shale interbeds

3 - Shoreline to regressive shallow marine facies successions 7 - Reef

3b - As above.but with incised non-marine fill 8 -. Deep marine carbonate apron

3c - As above,but growth faulted 9 - Carbonate slope.micrites and megabreccias

3d - Shoreline deposits-isolated by base-level drop 10 - Deep marine pelagic sediments

F i g u r e 14 Composite diagrams of l.r.teral faeces rslationships shuv,!, or, seismic oai< in a s'ope basin (one with deep water such ~s <

passive continental margin), and a ramp basin (one on the craton lacking very deep water). The depositional facies can be generally identified

by the overall lateral relationships, and by the large-scale features such as the slope, the mounds, and the reefs that can be imaged.

measuring a characteristic seismic of marine geological investigations of that the relative strengths of reflections

wavelet for an area, then mathematically the upper few hundred metres) are can be observed.

Reflection frequency

cally combining it with depth, velocity, recorded directly onto paper by analog (time interval between peaks) depends

and density data from sonic logs used processes. Multi-channel seismic data on the spacing of reflectors as well

to calculate acoustic impedance contrast (collected by the petroleum industry as processing. Reflection continuity

trasts. This procedure models or calculated for exploration and development) are depends on the lateral extent of the re-

lates from the sonic log data what the recorded digitally to allow subsequent reflector. Within a data set of relatively

seismic response of the rock units numerical processing by large computers under uniform conditions of acquisition, pro-

should be. Visual comparison of amplitude variations in

tude and frequency patterns on the a complex series of numerical operations - these characteristics may be inter-

synthetic seismogram can be made comparisons performed on the raw data. It is interpretable in terms of rock properties

with the real seismic record to estimate designed to enhance the reliability and sedimentary facies (see Seismic

depths (from log depths) to reflections (reduce multiple internal reflections Facies section below).

(Figs. 11, 13). Details of synthetic seismic and spurious reflections), sensitivity A typical seismic line through the

mogram construction are given in a (increase signal to noise ratio) and position - Venture gas field, Scotian Shelf, is

very understandable and readable siting (correct locations of reflections) shown in Figure 12.

Individual shot

form by Anstey (1982). A synthetic seismic from dipping beds) of the data. It points and traces cannot be seen on

seismogram is a very powerful tool for also allows the data to be displayed in this scale, but the more or less horizontal

calibrating the seismic response to the an interpretable fashion. Horizontal dark lines in Figure 12 are re-

stratigraphy of an area, because it Reflection amplitude, as mentioned reflections like those in the lower left of

shows which lithologic interval is represented above, is a function of acoustic impedance - Figure 11. The seismic line has been

sponsible for an individual reflection. pedance contrast. It is well enough annotated with formation names on

Single-channel seismic data (typical preserved during modern processing the left end, and ages on the right.

Figure 15 Upper Cretaceous and Tertiary section from the Venture area, Scotian Shelf. Erosional sequence boundaries (ESB) on seismic

lines are indicated by truncation or onlap of reflectors. Clinoforms are indicated by "C". Many rotational normal faults disrupt the clinoform sets

at the base of the Banquereau Formation. The ^ipJj^irnplitudejeflBctions in the Wyandot Formation originate from chalk beds which are

essentially analogous to a co^densed_section, deposited at the peak of the transgression when clastic input was low.

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These data are derived from the two interpreted to result from eustatic varia- some well-log cross sections.

wells, as tied in with synthetic seismo- tion of sea level, although other mech- Seismic reflections are in mos

grams. In this case, seismic facies se- anisms are now considered possible cases generated where sharp lithe

quences were defined on variations in (Chapter 1). However their origins are logic contrasts occur between succe:

amplitudes, frequencies, and continu- interpreted, these configurations are sive stratigraphic units. Lateral grad,'

ities of reflections. The growth faults in fundamentally important, large-scale tions of lithology or facies cannot t

the Jurassic section and the minor features seen on seismic lines and imaged. Sharp stratigraphic contr;.

fractures in the Lower Tertiary were delineated by correlating breaks in re- flections. Detailed parts of this line will be discussed later in the chapter.

SEISMIC STRATIGRAPHY

Seismic stratigraphy involves the application of seismic data to the study of regional and global sedimentary sequences and their bounding unconformities. It grew out of investigations performed by the Exxon Production Research Company, summarized initially in Payton (1977). This publication explains many of the fundamentals of large-scale stratigraphic analysis (both seismic and otherwise) and must be

understood by anyone attempting this kind of work. The interpretations of relative sea level changes purely in terms of eustasy (Vail et al., 1977) have recently been modified by some workers, but the basic principles of seismic stratigraphy are demonstrated in this publication (Payton, 1977).

Figures 11 and 12 emphasize a fundamental condition of seismic stratigraphy, namely that the scale and resolution of multichannel seismic data are very different from outcrop or well-log cross sections. Particularly at great depths where velocities are high, seismic data are generally incapable of resolving stratigraphic features less than 50 to 100 m in thickness. These differences in scale of resolution are emphasized in the example from the Mannville Group of Alberta (Fig. 11), where a gamma-ray log, a synthetic of the well, and a seismic line, show the differences in degree of vertical resolution. Recent detailed work using well-log cross sections (Figures 5 and 9) has revealed several unconformities within the Upper Mannville alone, but these all occur within one seismic sequence (definition in Chapter 1). Many recurring stratal patterns are seen on seismic lines that are useful for strati-

Figure 16 A core log from the Upper Jurassic-Lower Cretaceous Missisauga Formation graphic subdivision and analysis of Scotian Shelf, showing a regressive shoreface sandstone, followed by a transgressive sand

base-level changes. Figure 13 shows stone. The flooding surface occurs at the top of the oolitic carbonates (just above top o

many of *ne more important configura- mrp\ These transgressive carbonates generate high-amplitude, laterally continuous reflect

tions and this terminology applied to tions on seismic lines. The patterns of upwardly increasing reflection amplitude frpquerv

them. These configurations have been and continuity are shown in Figures 12 (curved arrows) and 17.

complex stacking of thalweg sandstones and overbank mudstones in large submarine fan valleys (Fig. 29 in Chapter 13). Similar seismic fades occur in nonmarine fluvial deposits with numerous variable-amplitude, discontinuities related to the presence of high-reflectivity coals and discontinuous shales in overbank areas, cut out by sandstone-filled channels. This seismic character differs from the much more continuous, more constant amplitude reflections generated from shoreline and shallow-marine elastics (Fig. 12), and from continuous basin floor turbidites (Fig. 31E in Chapter 13). High-contrast, high-amplitude reflections from interbedded shelf shales and limestones are easily separable in many examples from chaotic, almost tectonically free seismic facies. Some generalizations about seismic facies have been discussed by Sangree and Widmier (1977) and Roksandic (1978), and carbonate seismic facies by Bubba and Hatlelid (1977) and Sarg (1988).

Where uniform seismic data collection, processing, and display have occurred, simple qualitative calibration of reflection characteristics to core and well-log data from a specific area can also be made. This calibration procedure is analogous to the calibration and interpretation of SP or gamma-ray log patterns but is probably even more risky because of the lack of resolution of individual successions on the

seismic scale.

Figure 18 A seismic line with core and well-log control from the Upper Cretaceous Logan Canyon Formation, Scotian Shelf.

The core suggests interbedded open marine (from micropaleontology) and estuarine deposits, with erosion and generation of intraclasts and coal clasts. The alternation of open marine and estuarine deposits is interpreted to result from short-period base-level drops which cut erosion surfaces on the shelf (see Chapter 11). The seismic line shows a complete lack of seismic-reflection patterns (compare to Figures 12 and 17). The core is equivalent to the section imaged between 1.9 and 2.0 seconds.

Calibration of well logs, cores and shoreline clastic facies successions, by small clinoform sets (C in Fig. 17) of

seismic responses: Scotian Shelf with apparent transgressive shelf deposits, the basal, slightly deeper-water shelf

This calibration has been carried out as shown by many cores and deposits above. The patterns of

on two intervals of shallow-marine sediments (Fig. 17) from Venture gas field, upward-increasing reflection amplitudes

on the Scotian Shelf of eastern Canada. The transgressive units commonly are thin, discontinuous, and frequent

Canada. The Upper Jurassic to Lower Cretaceous capped by a thin bed of sandstone can be used to interpret the same style

Cretaceous Missisauga Formation the flooding surfaces of sedimentation in uncored areas.

(Fig. 16) consists mainly of stacked strong seismic reflections. These surfaces. The shaly Albian to Cenomanian

coarsening-upward shallow marine to faces are in many examples overlain Logan Canyon Formation shows com-

pletely different shallow-marine facies

in cores (Figs. 12, 18). The sharp-

based sandstones do not appear to be

parts of regressive coarsening-upward

successions; they are more randomly

interbedded. On the basis of their sedi-

mentary and biogenic structures in

cores, they can be interpreted as estu-

arine or even nonmarine deposits intercalated within the marine shales.

Although well control is not adequate to map incised valleys, it is believed that the basal surfaces of the sandstones represent erosional events resulting from short-term base-level drops.

The seismic response of this shallow-marine facies (Fig. 18) is completely different from that of the Missisauga interval (Fig. 12) discussed above. Amplitudes and times of arrival (depths) of individual reflections vary laterally in an irregular fashion across the interval. No vertical patterns of reflection amplitudes or frequencies can be documented.

Some sloping reflections with apparent onlaps exist (Fig. 18) which could be interpreted as delineating erosion surfaces, but resolution of these is poor. Small (less than 50 m high), laterally restricted clinoforms and minor growth faults also occur in places (Fig. 18).

The overall aspect of the seismic response of this interval is an absence of patterns of any seismic attributes, either vertically or horizontally. The discontinuous reflections have variable amplitudes which are not organized into any larger-scale patterns; this is due to lateral variation in the amount of sandstone, minor growth faults, and absence of major transgressive or condensed horizons. This seismic facies should be contrasted with the packages of upwardly increasing reflection amplitudes, frequencies and continuities shown in Figure 17.

Figure 21 Stratigraphy of the Mannville Group of the Alberta Basin from a proximal area (SE) to a distal one (NW). The lower part of the succession (up to the MFS, maximum LARGE-SCALE FACIES

flooding surface) consists of a transgressive systems tract, and the upper part is a highstand
RELATIONSHIPS

systems tract. The shale clinoforms are shown in Figure 8. Depositional environments from
Subsurface work has contributed a

outcrop and subsurface work are shown. great deal in recent years to knowl-

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edge of large-scale facies relation - tracts occur in the same arrangement

CONCLUSIONS

ships. It has particularly helped in understanding the way in which facies patterns differ from that of combined with allostratigraphy
and/

successions relate to one another and Posan, and Vail (1988). sequence stratigraphy, is now one of
the

to the different types of Large-scale subsurface analysis in most dynamic fields of
research in sedimentary geology.

Comparison of the most abbreviated form possible, a greater understanding of relationships volume with
previous editions of Facies

few of the most general conclusions between siliciclastic and carbonate Models shows the
increase in understanding of sedimentary systems achieved by subsurface study of these sediments in many areas. Figure 22

standing of sedimentary systems are

large-scale relationships. shows a cross section of one side of a the extent of the contribution of
subsurface

Large-scale facies associations can basin rimmed by carbonate banks and face work. While many
ideas about incised

be grouped into depositional systems reefs. The basin centre facies was defined. Facies have not
changed markedly;

systems tracts, depending on relationship - positioned beyond the bank edge, and in the last ten years,
ideas of facies relationships

ships to allostratigraphic bounding discontinuities consists dominantly of sandstones and relationships and the
nature and significance

continuities, positions in the basin with shales. The relationships between facies of the bounding
surfaces between

respect to other systems tracts, and these systems tracts imply that relationships - them have been
revolutionized.

whether they have overall progradational or retrogradational patterns transported across the carbonate
REFERENCES

(Fig. 19). Depositional systems tracts shelf, probably in incised valleys
Basic sources of information

are associated with specific bounding during periods of relative sea level fall.
discontinuities. For example, a transgressive At the slope break, they were defined.

whether they have overall progradational or retrogradational patterns transported across the carbonate

REFERENCES

(Fig. 19). Depositional systems tracts shelf, probably in incised valleys

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are associated with specific bounding during periods of relative sea level fall.

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gressive systems tract is capped by a posited as deltas, shorelines, or deep International Human Resources Develo

maximum flooding surface or con- sea fans, depending on the local basin ment Corporation, 168 p. densed section (Fig. 19), whereas a depth compared to sea level. During An exceptionally entertaining book e

highstand systems tract is capped by subsequent transgressions, some plaining the basics of the seismic metho

an unconformity (Fig. 19). T h e f a c i e s shelves were covered by thin siliciclas- Nonmathematical, for nonspecialists.

associations or systems tracts are tics but others c o m p l e t e l y lack this Asquith, G.B., 1982, Basic well-log analys

normally composed of several vertical facies. Rapid production of carbonate for geologists: American Association c

facies successions (Fig. 20). Individ- s e d i m e n t s during the relative rising Petroleum Geologists, Methods i

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ual s u c c e s s i o n s do not n e c e s s a r i l y sea level phase allowed aggradation

An excellent introduction to well-log an,

show the s a m e t r e n d . The a s s o c i a - a n d p r o g r a d a t i o n o f the c a r b o n a t e lysis, directed at the interpretation of tlun

tions of bounding discontinuities complex (Chapter 18). This kind of al- in rocks by use of mathematical equatioi

systems tracts have been reviewed by ternating highstand-lowstand deposi- and calibrations. Very clearly presented -

Posamentier and Vail (1988). These tion of c a r b o n a t e s and elastics has Payton, C.E., 1977, Seismic stratigraphy

authors interpret the organization of been termed reciprocal sedimen- applications to hydrocarbon explorative

d e p o s i t i o n a l s y s t e m s and erosional tation. Most well-established examples American Association of Petroleui

discontinuities almost entirely in terms have been imaged on seismic lines, Geologists, Memoir 26, 516 p.

of eustatic sea level variation. How- but well-log cross sections and rare A basic, necessary reference, no\

ever, other interpretations can be o u t c r o p s also s h o w t h e s e r e l a t i o n - somewhat dated by advances in th:

made in terms of tectonic subsidence ships. field.

and variations in the rate of sedimen-

tation. An example of this is shown in

Figure 2 1 , from the Lower Cretaceous

Mannville Group of Alberta. The Lower

to Middle Mannville (below MFS

in F i g . 2 1) c o m p r i s e s a l o w s t a n d

to transgressive systems tract made up of individual progradational facies successions. The thin shale directly above the Bluesky Sandstone contains the maximum flooding surface, as shown by regional cross sections. Figure 22 Cross section illustrating a succession that results from "reciprocal" sedimentation. The Upper Mannville is a strongly tilted. Note that carbonate platforms and bank edges intertongue basinward with elastics, progradational highstand depositional commonly sandstones proximally and shales distally. This configuration of lithofacies is in systems tract. The stratigraphy of this interpreted to result from a carbonate shelf prograding and aggrading under conditions of rising relative sea level. During a base-level drop, the elastics are transported through clastic wedge has been interpreted as breaks and channels in the carbonate margin, and deposited either as turbidites and other the result of variations in tectonic sub-deep sea deposits, or in other situations as coarse-grained deltas. Because of the low relative sea level, carbonate deposition is minimal at that time. Thus relative sea level fluctuations from the Cordilleran "orogenic" results in deposition of lithified carbonates alternating with low-relief. basin, rather than eustatic variations in sea level. The cross section is a composite from the Permian Guadeloupe P. complex sea level. The Mannville systems west Texas, and seismic lines across the northwest Australian shelf.

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